



ISO 22004:2014



For Valero Trust Company for Trading and General Supplies

Food safety management systems – Guidance on the application of ISO 22004



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0 Introduction

0.1 General

This International Standard provides guidance on the use of ISO 22000 in four parts: planning, implementation, verification and improvement (Plan, Do, Check, Act). The clauses of ISO 22000 link together to form a food safety management system (FSMS).

Correspondence with the relevant clauses of ISO 22000 is given in the headings as well as in [Annex A](#) (see [Tables A.1](#) and [A.2](#)).

0.2 Food chain approach

Establishing an FSMS is a tool used to mitigate the risk to public health associated with the organization's products; it is also useful for ensuring compliance with statutory/regulatory requirements and/or those specified by customers.

ISO 22000 promotes the adoption of a systematic approach for developing, documenting, implementing and maintaining an FSMS. Integral to this approach are supply chain management (supplier evaluation and approval) and ensuring the safety of products during distribution.

0.3 Process approach

ISO 22000 also follows the "process approach" (i.e. management of a system of interrelated processes with identified interactions).

An advantage of the process approach is the ongoing control it provides between the individual processes within the system.

When used within an FSMS, the process approach emphasizes the importance of:

- a) understanding and fulfilling the ISO 22000 requirements;
- b) considering food safety as a process;
- c) considering traceability as a process;
- d) monitoring of process performance and effectiveness;
- e) continual improvement of processes based on objective measurement(s).

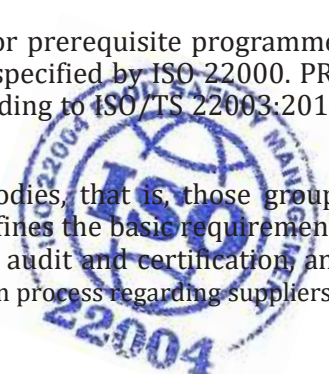
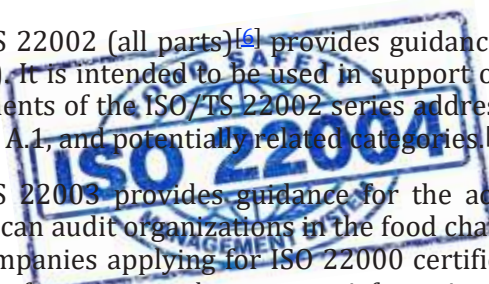
Any and all parties, as defined by internal and external communication, can play a role in defining process requirements. Evaluating the satisfaction of these entities requires the collection and analysis of information to determine whether or not the organization has been able to meet these demands.

0.4 The ISO 22000-related documents

The ISO 22000 family of documents comprises of a number of individual International Standards and Technical Specifications, which are interrelated and supplementary to each other (see [Figure 1](#)). ISO 22000 is the primary International Standard, which defines the requirements and to which the other International Standards and Technical Specifications within the family are linked.

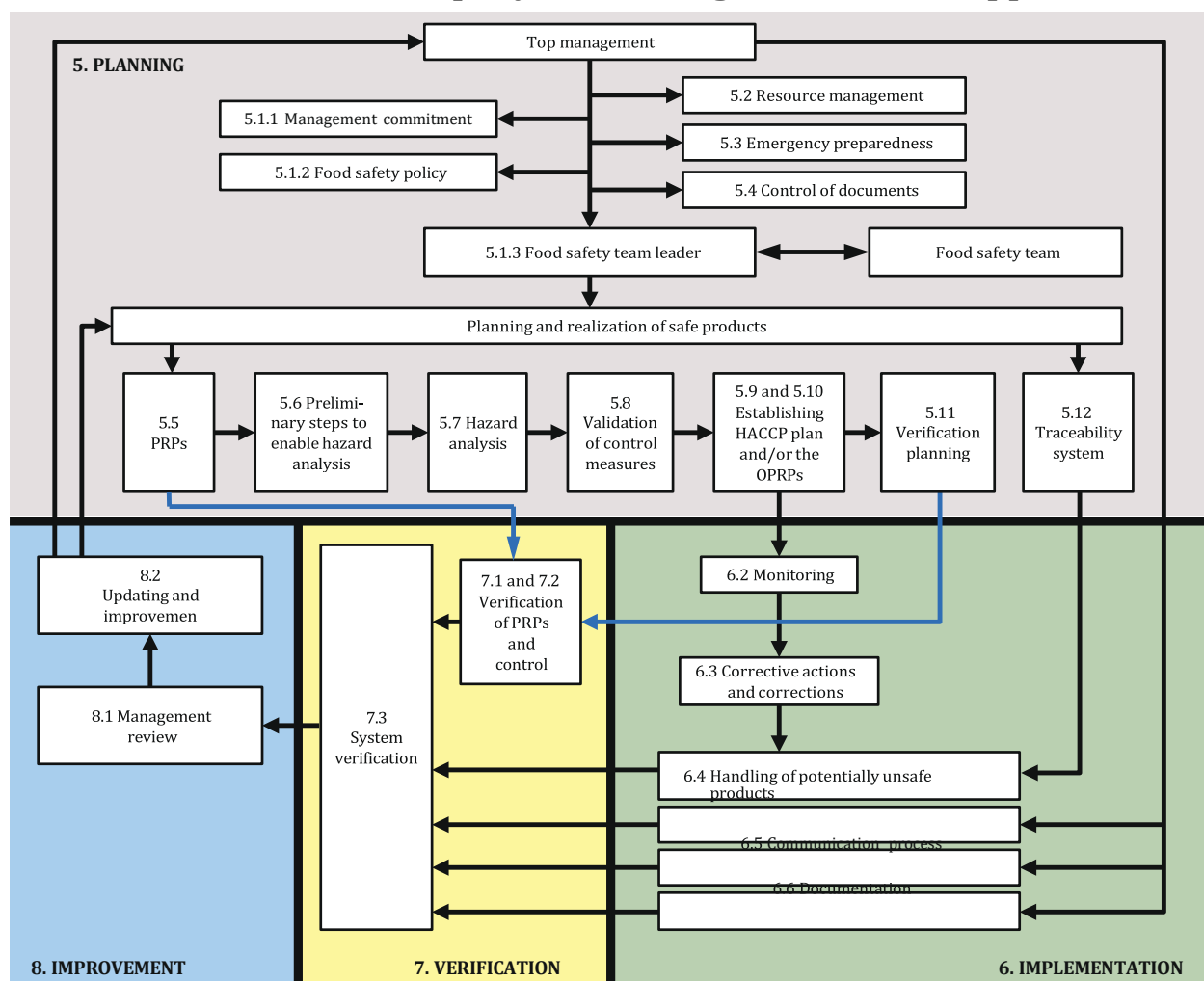
ISO/TS 22002 (all parts)^[6] provides guidance to meet the requirements for prerequisite programmes (PRPs). It is intended to be used in support of the requirements for PRPs specified by ISO 22000. PRP documents of the ISO/TS 22002 series address food chain categories according to ISO/TS 22003:2013, Clause A.1, and potentially related categories.^[7]

ISO/TS 22003 provides guidance for the accreditation of certification bodies, that is, those groups which can audit organizations in the food chain under ISO 22000. It also defines the basic requirements for companies applying for ISO 22000 certification, the rules applicable to audit and certification, and provides for customers the necessary information and confidence in the certification process regarding suppliers.





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NOTE The numbering refers to the (sub)clauses of this International Standard.

Figure 2 — Illustration of the PDCA approach to FSMS

Development, maintenance and improvement of the system are addressed by planning, monitoring, verification and updating as shown in [Figure 2](#). Maintenance and improvement of the system is addressed through a number of cycles of planning, validation, monitoring, verification and updating. Within an operating system, changes can be initiated at any of these phases.

PRPs are established to ensure that basic requirements and activities are in place, thus ensuring the establishment of a hygienic environment. Additionally, some PRPs are practices globally recognized as contributing to the provision of safe and suitable [\[13\]](#) food.

Significant hazards identified during hazard analysis are controlled by identified control measures that are subsequently categorized into two categories of control measures (given in 7.4.4 of ISO 22000:2005):

- control measures applied at CCPs: these control measures are managed by a HACCP plan. Such control measures have defined critical limits that can separate acceptable product from potentially unacceptable (unsafe) product. In addition, their implementation can be monitored in a manner that enables detection of any loss of control within a timeframe sufficient to effectively control affected product. Failure to meet critical limits will result in a potentially unsafe product.
- control measures applied as OPRPs: these control measures cannot be managed by an HACCP plan but can be managed by similar means such as an OPRP plan. In the case of OPRPs, such control measures do not have a critical limit, but should have an action limit or an action criterion, which



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demonstrates that the OPRP is in control. Failure to meet these established criteria will lead to corrective actions.

The affected products “shall be evaluated with respect to the cause(s) of the nonconformity and to the consequences thereof in terms of food safety and shall, where necessary, be handled in accordance with 7.10.3” (of ISO 22000:2005).

NOTE The traditional Codex decision tree does not include the concept of OPRP.

Control measures applied at CCPs and as OPRPs are intended to control all the hazards assessed as significant during hazard analysis. Control measures or combinations of control measures applied at CCPs and as OPRPs should be validated to ensure that they are effective. Validation is not required for PRPs.

Table 1 — Actions linked to PRPs, OPRPs and CCPs, as well as the relationship between them

PRP ^a	OPRP	CCP
Preliminary development based on experience and assessment of reference documents.	Identified once PRPs are developed, and based on the hazard analysis taking the PRPs into account. OPRPs and CCPs are product and/or process specific.	
Measures impacting food suitability ^[13] and safety.	Measures (or combination of measures) controlling hazards that are remaining after the implementation of PRPs.	
General measures that are not specific to any hazard.	Measures (or combination of measures) that are specific to each hazard or group of hazards.	
Measures related to creating the environment for the production of safe food.	Environment- and/or product-related measures (or combination of measures) to prevent contamination or to prevent, eliminate or reduce hazards to an acceptable level in the end product.	
Monitoring where relevant and feasible ^b .	Validation	
	Measurable or observable action criterion(a) or action limit(s) demonstrating that the OPRP is in control.	Measurable critical limit(s) is (are) demonstrating that the control measures (or combination of measures) associated with each CCP is in control.
	Monitoring of the implementation of control measures (or combination of measures).	
	Corrective actions, and/or corrections as needed.	Corrective actions, and/or corrections.
	Monitoring records kept.	
Scheduled verification of implementation	Scheduled verification of implementation. Verification of achievement of planned hazard controls.	
^a PRPs are often referred to as good practices, such as good hygienic practices, good agricultural practices and good manufacturing practices.		
^b Many PRPs do not lend themselves to monitoring. Thus monitoring activities should be in place where feasible.		

External competences may be used by the organization to develop the combination of PRPs and control measures, provided they meet the requirements of 7.2 to 7.8 of ISO 22000:2005 (see [Clause 9](#)).





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5 Planning

5.1 Management responsibility (Clause 5 of ISO 22000:2005)

5.1.1 Management commitment (5.1 of ISO 22000:2005)

The organization's FSMS should be based on the key elements of food safety described in the Introduction and Scope of ISO 22000:2005.

It is intended that top management allows enough time to review the management system. It is the responsibility of top management to ensure that the review includes an assessment of performance against the food safety policy and the related objectives that have been set.

The organization can achieve sustained success by consistently meeting the needs and expectations of its interested parties.

To achieve sustained success its top management should:

- have a long-term planning perspective (applying the continual improvement principle);
- list stakeholders involved with external and internal communication, assess their individual potential impacts on the organization's performance, as well as determining how to meet their needs and expectations in a balanced way;
- engage stakeholders and keep them informed of the organization's activities and plans;
- identify associated short- and long-term risks and implement an overall strategy for the organization to mitigate them;
- regularly assess compliance with current plans and procedures, and take appropriate corrective actions and update the programmes as required.

5.1.2 Food safety policy (5.2 and 5.8.1 of ISO 22000:2005)

Top management should establish the organization's food safety policy, so that the mission, vision and values are accepted by relevant interested parties. The organization's policies should be reviewed on a regular schedule and revised as needed.

The organization's food safety policy should incorporate activities needed to achieve food safety statutory/regulatory requirements, and the food safety requirements of customers.

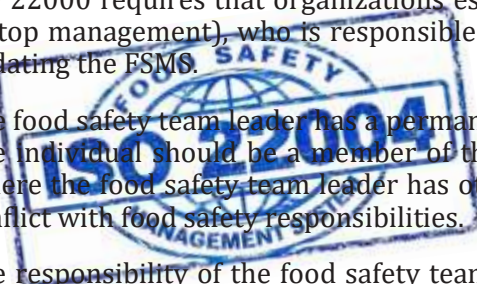
Programmes should be developed, documented, implemented, maintained and updated, in such a way as to enable the continuous review of the performance regarding the FSMS, and allowing for any necessary improvements.

5.1.3 Food safety team leader (5.5 of ISO 22000:2005)

ISO 22000 requires that organizations establish a food safety team and designate a leader (nominated by top management), who is responsible for developing, documenting, implementing, maintaining and updating the FSMS.

The food safety team leader has a permanent function, which is central to the FSMS of any organization. The individual should be a member of the organization and should understand its food safety issues. Where the food safety team leader has other responsibilities within the organization, these should not conflict with food safety responsibilities.

The responsibility of the food safety team leader may include liaison with external parties on matters relating to the FSMS (e.g. chamber of commerce, consultant, inspection agencies, suppliers, customers).





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It is recommended that the food safety team leader has extensive knowledge of hygiene, food safety management and application of the HACCP principles.

5.2 Resource management (Clause 6 of ISO 22000:2005)

5.2.1 Provision of resources (6.1, 6.3 and 6.4 of ISO 22000:2005)

The organization should ensure resources (such as equipment, facilities, materials, energy, knowledge, budget and personnel) are used effectively. A process, which provides, allocates, monitors, evaluates, optimizes and maintains those resources, should be in place.

The organization should periodically review the availability and suitability of the identified resources, including outsourced resources, and take action as necessary. The results of these reviews should also be used as inputs into the organization's reviews of its strategy, objectives and plans.

5.2.2 Training (6.2.2.1 and 6.2.2.2 of ISO 22000:2005)

Personnel are a most valuable and critical resource. It is necessary to ensure that their work environment encourages personal growth, learning, knowledge transfer and teamwork (see [5.2.1](#)). The organization should ensure that the personnel understand the key elements of an FSMS (e.g. HACCP, PRP, management) and the importance of their contribution and roles in the FSMS.

Personnel involved in food safety (see 6.2.1 and 6.2.2 of ISO 22000:2005) need to be trained, training should be documented. In addition, organizations in the food chain should establish protocols that the training is effective and that the workers being trained demonstrate that they are competent (see [Annex C](#)).

5.3 Emergency preparedness (5.7 of ISO 22000:2005)

The organization should be aware of potential emergency situations, including but not limited to, natural disasters, environmental accidents, bioterrorism, or other incidents that can affect food safety and/or production.

Response is dependent on local or national regulatory expectations and may include communication about the crisis both within and outside the organization. This would include communication with relevant stakeholders such as media, customers, employees, law enforcement and governmental agencies (see [6.5](#)).

An example of verification method is a mock/test exercise for emergencies that could be conducted to assess response and business continuity plans (see [5.11](#)). Effective traceability and product withdrawal/recall procedure(s) is (are) essential (see [5.12](#)).

NOTE ISO/TS 22002-1:2009, Clause 18 specifies details relating to food defence, biovigilance and bioterrorism.

5.4 Control of documents including records (4.2.2 and 4.2.3 of ISO 22000:2005)

Document control is a basic management programme to ensure that only approved, up-to-date documentation (see [6.6](#)) is available and used throughout the organization. Inadvertent use of out-of-date documents can have significant negative consequences on food safety, costs and customer satisfaction.

The organization should implement a written procedure to establish a document control programme. The document control programme should include the following elements:

- how documents are approved prior to issue;
- how documents are amended, updated and re-approved, including disposal of outdated and obsolete documents;





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- c) how to identify the current revision status of, and changes to, documents (e.g. by date or issue number) and by highlighting changes made to the previous version;
- d) how it is ensured that documents are available where needed (storage and accessibility policy);
- e) how the distribution of documents is controlled. The site may maintain a master list of documents;
- f) how documents of external origin, needed for the FSMS' planning, operation and assessing, are identified, controlled and distributed in accordance with list item e);
- g) a records control procedure specifying which records are kept, how they are stored, retrieved, identified and destroyed.

Document/record control and document/record management programmes are intended to conform to applicable government regulations.

5.5 PRPs (7.2 of ISO 22000:2005)

Prerequisite programmes (PRPs) are basic programmes essential for maintaining a hygienic environment for production, processing and/or handling of product and are not implemented for the purpose of controlling specific identified hazards. ISO/TS 22002 (all parts)^[6] provides guidance on the selection of PRPs applicable to different sectors/categories of the food chain. Other sources for selecting appropriate PRPs are listed in Annex C of ISO 22000. The organization should design the PRPs so that they are appropriate to the type of food, premise, facilities, process, location in the food chain, etc.

While PRPs and control measures are implemented and applied simultaneously in the organization, they should be developed sequentially. PRPs provide the environment for the safe production of products. This explains why the PRPs should be established and designed before proceeding to the hazard analysis. However, planning, implementation and updating of the PRPs and other parts of the FSMS may trigger the need for changes or improvements to the PRPs.

The implementation of PRPs should be verified.

5.6 Preliminary steps to enable hazard analysis (7.3 and 7.7 of ISO 22000:2005)

All relevant data and information needed to conduct a complete hazard analysis should be gathered and documented. This includes identifying and assessing hazard occurrence in the raw materials, ingredients, processing, storage/handling environments and in the end products. Operations should be assessed to identify areas for potential cross contamination. Products may be grouped into similar categories depending on ingredients, processes and/or hazards.

The food safety team should also describe the intended use of the end product and consumer categories. Where appropriate, part of this process is to determine the consumption mode and to determine whether or not instructions for the proper preparation or use of the product are included in the label or otherwise communicated.

When establishing shelf-life, the food safety team should investigate:

- the transportation and storage conditions;
- information of actual use of the product (e.g. how the end product is cooked before consumption or how the food is intended to be used by another organization in the food chain);
- reasonably expected handling and mishandling of the end product by the end-user (e.g. subsequent processing or handling by the consumer including at the consumer's home).

All relevant data and information that are part of the hazard analysis should be maintained and updated as needed.





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5.7 Hazard analysis (7.4 of ISO 22000:2005)

5.7.1 General

The process of hazard analysis is illustrated in [Figure 3](#).

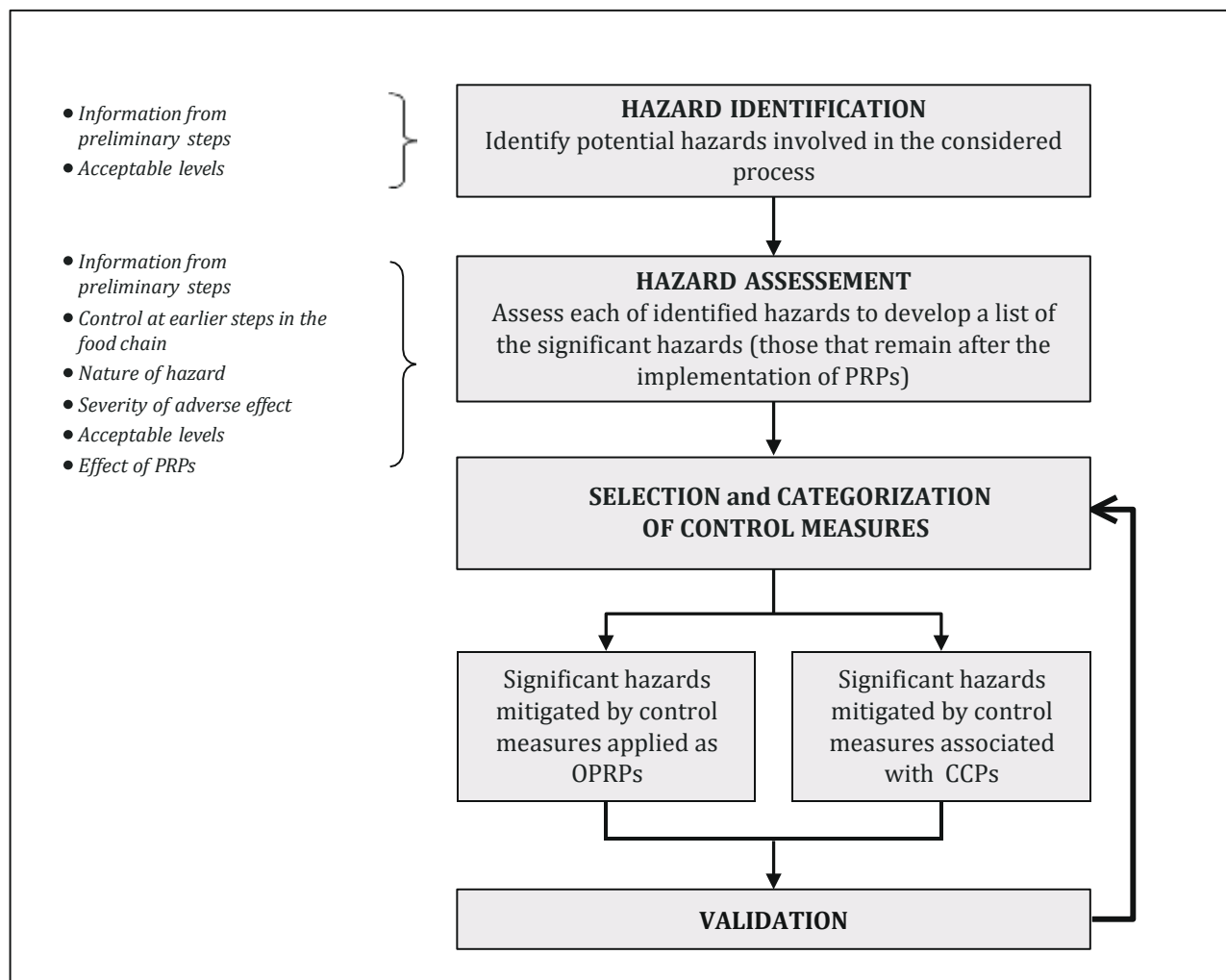


Figure 3 — Illustration of the hazard analysis process

5.7.2 Hazard identification and determination of acceptable levels (7.4.2 of ISO 22000:2005)

According to ISO 22000, for each significant hazard, the organization shall establish acceptance levels¹⁾ for the end product(s). Acceptance levels relate to the hazards in question. Determination of acceptance levels may be based on multiple sources such as:

- criteria established by relevant statutory/regulatory authorities;
- food safety specifications²⁾ or other information provided by buyers, in particular for end products intended for further processing or use other than direct consumption;
- in-house or external data collected by the food safety team related to specific hazards.

1) The “acceptable level” means level of a hazard in the end product that shall not be exceeded in order to ensure food safety.

2) According to Codex Alimentarius, such specifications or levels can be expressed as Performance objectives.





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5.7.3 Hazard assessment (7.4.3 of ISO 22000:2005)

The role of hazard assessment is to determine each of the hazards identified above in [5.7.2](#) in order to specify significant hazards. In conducting the hazard assessment, the following should be taken into consideration:

- a) the source(s) of the hazard (e.g. where and how it can be introduced into the product and/or its environment);
- b) the probability or likelihood of occurrence of the hazard (such as frequency of occurrence at the highest possible levels and/or statistical distribution of levels). Consideration should be given to steps preceding and following the specified operation within the same system, the process equipment, service activities and surroundings, as well as to the preceding and following links in the food chain. In addition, consideration should be given to areas of possible cross contamination;
- c) the nature of the hazard (e.g. ability to multiply, deteriorate the food and/or produce toxins);
- d) the severity of the adverse health effects which can be caused by the hazard, with consideration to vulnerable population;
- e) the acceptable level in the end product taking into account measures taken at subsequent steps in the food chain (e.g. further processing, transportation, distribution and consumption).

The hazard analysis is performed to establish which hazards are significant and, therefore, subject to validated control measures. It may occur that identified hazards are not determined to be significant: for example, when an identified hazard is within acceptable levels without further intervention by the organization.

If the information required for conducting the hazard assessment is unavailable to the food safety team or organization, additional information can be obtained from scientific literature, databases, regulatory agencies, or industry experts.

A list of those hazards that have been identified as significant and need to be controlled by OPRPs or at CCPs should be established. Where no such hazards can be identified, the PRPs in place are effective and sufficient to achieve safe products.

5.7.4 Selection and assessment of control measures (7.4.4 of ISO 22000:2005)

An appropriate combination of control measures should be selected which are capable of preventing, eliminating or reducing to acceptable levels those food safety hazards identified as significant.

Guidance on categorizing control measures is provided in [Table 1](#). See also the Note to [5.9.2](#).

5.8 Validation of control measures or combination of control measures (8.2 of ISO 22000:2005)

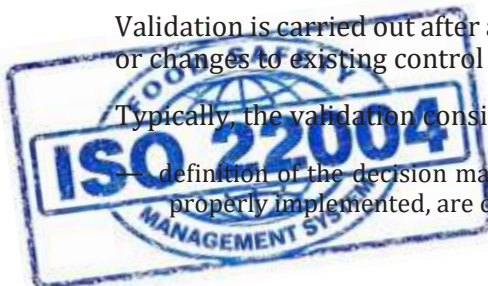
Validation is the obtaining of evidence, demonstrating that the control measures selected above in [5.7.4](#), managed by the HACCP plan and by the OPRPs are capable of controlling the identified hazards (see [6.1](#) and References [\[12\]](#) and [\[16\]](#)).

The validation process measures and/or assesses the control measures, or combination of control measures, to determine that they are capable of achieving the intended level of hazard control.

Validation is carried out after any hazard analysis, and prior to implementation of new control measures, or changes to existing control measures.

Typically, the validation consists of the following:

- definition of the decision making parameters and criteria that will demonstrate that the control measures, if properly implemented, are capable of consistently controlling the hazard to the specified outcome;





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- collection of pertinent documentation needed to the validation;
- conduct of the validation studies;
- analysis of validation results;
- documentation of the validation study.

Different methods may be used to validate control measures. It is imperative that the most appropriate validation method be selected. If the expertise is not available within the organization, the use of external expertise will be needed (see also [Clause 9](#)). Validation studies should be conducted before the actual food process is operated for the first time. Validation methods include, but need not be limited to:

- a) reference to scientific or technical literature;
- b) previous validation studies, or historical knowledge of the performance of the control measure(s);
- c) scientifically valid experimental data for example to demonstrate the adequacy of control measure(s);
- d) collection of data during operation of the entire food production process;
- e) mathematical modelling: predictive microbiology where appropriate, for example, to validate shelf-life (including reasonably foreseeable time/temperature variation) or to determine alternative time/temperature combinations for heat treatment or conditions for no heat treatment. Mathematical models should only be used if they have been validated using actual processing conditions and product characteristics (e.g. a_w , pH);
- f) surveys, which should be developed by experts. Sample size and individual selection for the survey should be established in accordance with proven survey techniques.

If relying upon validations carried out by others, care should be taken to ensure that the conditions of the intended application are consistent with those identified in the referenced validations.

For some well-established processes (e.g. time and temperature combinations for milk pasteurization), it may be sufficient to gather only the data on the conditions or attributes specific to the operation in question.

Validation data should be reviewed following new scientific or statutory/regulatory information, a system failure, or any significant change to the processes or products. Changes to products or processes include changes in raw material or ingredients, in a process step, in organizational management, or changes implemented in response to a critical limit, or in the conditions of use by end-consumers, etc.

5.9 Establishing the HACCP plan (7.6 of ISO 22000:2005)

5.9.1 HACCP plan (7.6.1 of ISO 22000:2005)

The HACCP plan is a controlled document (see [6.6.2](#)). Among its many parts are descriptions of how the control measures associated with CCPs are managed. ISO 22000 specifies the requirements for the content of the HACCP plan.

5.9.2 Identification of critical control points (7.6.2 of ISO 22000:2005)

A CCP is a step in the process at which control can be applied, essential to prevent or eliminate a food safety hazard, or reduce it to an acceptable level. Control measures associated with CCP will meet the following criteria:

- critical limits need to be established that differentiate an acceptable product from a potentially unacceptable product. Failure to meet critical limits will result in potentially unsafe product.





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- critical limits can be monitored in a timely manner so that action(s) can be taken to ensure foods will meet acceptable levels of hazards in the end product.

NOTE In practice, control measures associated with CCPs are identified during the categorization of control measures (see [Clause 4](#)).

5.9.3 Determination of critical limits for critical control points (7.6.3 of ISO 22000:2005)

The critical limits have to be achievable as part of normal process operations and be validated as part of the validation described in [5.8](#).

For CCPs intended to control more than one hazard, the critical limit(s) should be determined relative to each hazard, and the most stringent limit applied.

5.9.4 System for the monitoring of critical control points (7.6.4 of ISO 22000:2005)

The control measures applied at CCPs are typically monitored through chemical or physical measurements such as pH, time, temperature etc. The results of monitoring should be recorded.

The frequency of monitoring is determined by the organization, to enable it to detect any loss of control within a specified time frame sufficient to control the affected product. Critical limits may be monitored continuously or intermittently.

5.9.5 Actions when monitoring results exceed critical limits (7.6.5 of ISO 22000:2005)

Exceeding critical limits means that the products are potentially unsafe. Both corrections and corrective actions should be planned, documented where possible, and applied when necessary (see [6.2](#) and [6.3](#)).

5.10 Establishing the OPRPs (7.5 of ISO 22000:2005)

5.10.1 OPRP plan

ISO 22000 specifies the requirements establishing and managing OPRPs. This can be organized through an OPRP plan [see Note to item a) [Clause 4](#)].

5.10.2 Monitoring of OPRPs

To meet the requirements for monitoring of OPRPs, the organization should establish appropriate action limits or action criteria.

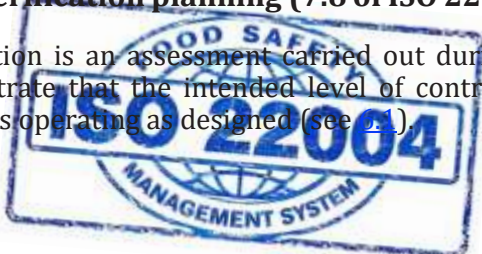
These limits should correspond to any operational parameters used in validation as described in [5.8](#). Where limits cannot be determined due to the nature of the measure, it is the actual application of the specific procedure that is to be monitored. Monitoring frequency should be established by the organization.

5.10.3 Actions when monitoring results exceed action limits or action criteria

When monitoring results exceed action limits or action criteria, corrective actions and, where appropriate, corrections should be planned and applied (see [6.2](#) and [6.3](#)).

5.11 Verification planning (7.8 of ISO 22000:2005)

Verification is an assessment carried out during or after the operation. The role of verification is to demonstrate that the intended level of control has actually been achieved and that the food safety system is operating as designed (see [6.1](#)).





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Care should be taken to take full advantage of any statutory/regulatory reporting, or early warning systems of noted or perceived hazards.

Labelling is, in its broadest sense, a tool for communication with customers/consumers.

6.5.3 Internal communication (5.6.2 of ISO 22000:2005)

The internal communication system of the organization should ensure that sufficient relevant information and data are available to all personnel involved in the various operations and procedures. The food safety team leader has a major role in the internal communication of food safety issues within the organization.

Any member of the organization observing an activity which may have an impact on food safety should know how to report the issue and to whom.

6.6 Documentation (4.2 of ISO 22000:2005)

6.6.1 General

ISO 22000 requires that the FSMS be documented (given in 4.1 of ISO 22000:2005) and that the organization establishes a document control programme (see [5.4](#)).

A document is a paper or any other media providing information.

ISO 22000 refers to a documented procedure or statement.

Documentation should be a value-adding activity to the benefit of the organization.

Documentation consists of:

- documents used to document the FSMS, such as policy statements, PRP, HACCP plan, verification plan, communication plan, and traceability system;
- documents describing how to carry out an activity and/or intended input or outcome, such as procedures, work instructions and specifications;
- records providing evidence that an activity has been performed or results have been achieved.

6.6.2 Documents used to document the FSMS

The type and extent of system documentation vary from one organization to another. This is due to the size and complexity of the activity, the skills of personnel, OPRPs and HACCP plans.

All documents relevant for food safety, whether developed externally or in-house, should be a part of the FSMS.

NOTE Additional guidance on use of consultancy and externally developed generic guidelines is provided in [Clause 9](#).

Documents include, but are not limited to, product specifications, HACCP plan, OPRPs and PRPs, and contracts for any outsourced processes (e.g. pest control, product testing). The documents used by the organization should be available when and where required and may be in any approved format (e.g. paper, electronic or picture).

All documents relevant for food safety should be accessible to the involved personnel.

Documentation required by the standard to document the FSMS includes:

- a) food safety policy (given in 4.2.1 of ISO 22000:2005);
- b) list of measurable food safety related objectives (given in 4.2.1 of ISO 22000:2005);

